Natalyx MVP Fixes

Consolidated plan overview - fourteen coordinated work packages

L. Rhodes (plan owner) - A. Zhao (technical feedback reconciled)

2026-07-12

Contents

1 Purpose and approved planning update **3**

2 Current position **3**

3 Architecture spine **4**

4 Work-package map **4**

5 Suggested ownership and reciprocal review **5**

6 Delivery sequence **5**

7 Programme gates **6**

8 Recommendation disposition matrix **7**

9 Work-package details **14**

9.1 WP00: Preservation, cloud diagnosis, and PR reconciliation 14

9.2 WP01: CI and PostgreSQL evidence gate 15

9.3 WP02: Versioned migrations and readiness 16

9.4 WP03: Authentication, invitations, and resume 17

9.5 WP04: Node 0 workspace and paid-case separation 18

9.6 WP05: Node 0 Free, Pathfinder, and resume 19

9.7 WP06: Node 0 Complete and GC soft onboarding 20

9.8 WP07: Commercial configuration, enrollment, and promotion 21

9.9 WP08: Payments, invoices, credits, and refunds 22

9.10 WP09: Documents, active case, and provider handoffs 23

9.11 WP10: Intelligence, memory, proposals, and Ask safety 25

9.12 WP11: Learn release and design-system contract 26

9.13 WP12: Ask/Learn shell and mobile cutover 27

9.14 WP13: Deployment, cloud gates, and PHI readiness 28

10 Founder decision resolution and remaining controls **30**

10.1 Immediate Alan decisions 36

10.2 Later Alan decisions 36

10.3 External dependencies 37

1 Purpose and approved planning update

This is the reconciled planning document for cleaning up the current Natalyx MVP delivery surface without losing useful work or compounding the present architecture. It preserves fourteen bounded work packages and now contains 52 suggested pull requests. The detailed Markdown and PDF plan for each work package is supplied separately in the `MVP Fixes` Drive folder.

The GitHub Project is deliberately benched until this plan is reviewed. Only PRs whose local-preparation field expressly permits isolated, uncommitted, synthetic-only preparation may be touched before joint acceptance; nothing authorizes commit, PR creation, merge, branch cleanup, cloud changes, payment activation, real-user use or PHI processing.

Alan's completed technical feedback, corrected #122 and preserved Wave 2 have been independently reconciled. The founder interview is complete: all 24 decision IDs are classified, founder outcomes are final where recorded, and remaining technical controls and external approvals are explicitly separated from those final policies.

2 Current position

Area	Current conclusion	Planning consequence
Live dev	The inspected endpoint returned ALB 503.	Treat live recovery and known-revision evidence as Gate A0, separate from green CI.
PR 122	Authoritative baseline for named invitation/resume/lens behavior and tests, but not the final schema/domain architecture.	Selectively port behavior/tests; supersede runtime DDL and conflated access/provenance models.
PR 124	Alan's complete review confirms high extraction value but a stale/conflicted and unsafe merge unit.	Fully supersede as a merge unit; port only approved concepts/tests/copy/assets.
Wave 2	Exact three-commit stack at 5b63e3a; 535 tests passed locally, with restart-loss and stale-review defects independently reproduced.	Preserve paused; adapt phase-1 presentation, transaction patterns and tests under P03/P09/P13.
Access / CI	Identity/participation/access were conflated and RLS absent. GitHub Team is commercially approved but not yet activated or configured.	P03-05 resolves the access model; P13-05 precedes B1; D-CI-01 keeps A0 and implementation merges blocked until enforcement is active and verified.
Node 0	Free exploration, Complete entitlement, commitment, payment and paid case are currently too entangled.	Separate their entities and state machines before expanding UI.
Commercial	Founder beta pricing, commitment, credit, refund, loss and grace policies are final; no append-only money ledger or promotion authority exists yet.	Implement the settled policies as versioned processor-neutral contracts; external terms/tax review still precedes live acceptance or charging.

Area	Current conclusion	Planning consequence
Documents/PHI	Documents are metadata-only and the provider branch is unreviewed.	Keep synthetic-only through A2, approved PII at B1, and PHI blocked until formal B2.

3 Architecture spine

- Node 0 begins as a free workspace; opening it never creates a canonical paid case.
- Workspace, entitlement, commercial commitment, payment, case, invitation/claim, document and journey-stage lifecycles remain separate.
- Identity binds principal to person; participation records the domain relationship; explicit resource grants carry capabilities; application and RLS authorization also consider ownership and purpose.
- Only `byos.workspace_promotion_authorized.v1` authorizes exactly one `enrollment_pending` case; browser redirects and raw processor callbacks never do.
- IP observations remain IP-attributed and provisional. GC confirmation appends a GC-authored revision; it never silently overwrites or upgrades the source.
- Metadata requirements/readiness/work items are not document content. Content versions have a separate upload, quarantine, scan, review, replacement and download lifecycle.
- The app owns durable case truth, authorization and PHI policy. Intelligence returns advisory proposals only. Learn artifacts and design-system releases remain pinned repository contracts.
- PostgreSQL is production authority. Versioned, ordered migrations replace runtime DDL, with locking, checksums, expand/contract compatibility and forward repair.
- Forced RLS and a non-owner runtime precede B1 real-user PII. PHI remains disabled until B2 evidence and human acceptance; merge or deployment success accepts neither gate.

4 Work-package map

WP	Work package	Suggested PRs	Proposed lead	Reviewer	Gate range
WP00	Preservation, cloud diagnosis, and PR reconciliation	1	Luke	Alan	Pre-A0
WP01	CI and PostgreSQL evidence gate	1	Alan	Luke	A0
WP02	Versioned migrations and readiness	4	Alan	Luke	A0, A1
WP03	Authentication, invitations, and resume	5	Alan	Luke	A1, A2
WP04	Node 0 workspace and paid-case separation	3	Alan	Luke	A1
WP05	Node 0 Free, Pathfinder, and resume	3	Luke	Alan	A1
WP06	Node 0 Complete and GC soft onboarding	3	Alan	Luke	A2

WP	Work package	Suggested PRs	Proposed lead	Reviewer	Gate range
WP07	Commercial configuration, enrollment, and promotion	4	Luke	Alan	A2
WP08	Payments, invoices, credits, and refunds	4	Luke	Alan	A2
WP09	Documents, active case, and provider handoffs	6	Alan	Luke	A2
WP10	Intelligence, memory, proposals, and Ask safety	4	Alan	Luke	A2
WP11	Learn release and design-system contract	4	Luke	Alan	A1, A2
WP12	Ask/Learn shell and mobile cutover	3	Luke	Alan	A2, B1
WP13	Deployment, cloud gates, and PHI readiness	7	Alan	Luke	A0, A1, A2, B1

5 Suggested ownership and reciprocal review

This is the proposed working split for discussion, chosen to keep product/commercial/content-heavy work with Luke and foundation/security/data/runtime work with Alan while retaining reciprocal review:

Lead	Work packages	Primary responsibility	Review partner
Luke	WP00, WP05, WP07, WP08, WP11, WP12	Programme reconciliation; Node 0 Free product boundary; commercial/payment contracts; Learn/design-system release; participant shell/cutover.	Alan reviews technical architecture, migration, security and failure behavior.
Alan	WP01, WP02, WP03, WP04, WP06, WP09, WP10, WP13	CI/PostgreSQL; migrations; auth/invitations; workspace/case model; GC onboarding; documents/provider handoff; intelligence; deployment/gates.	Luke reviews product, privacy, commercial, role, truth and release acceptance.

External security, privacy, payment, finance, legal, accessibility or provider review remains mandatory where the detailed package names it. The split can be amended before implementation; it is not an assignment imposed by the plan.

6 Delivery sequence

Phase	Work	Can run in parallel	Exit condition
Planning closeout	Founder decisions incorporated; WP00 is ready for Alan review and joint acceptance	Only expressly marked isolated, uncommitted preparation	Canonical Markdown/PDF package is validated and Drive readback succeeds.
Foundation	WP01, WP02 and A0 portions of WP13	Learn/DS source work in WP11	Current tests run on PostgreSQL; migration ADR/runner and known-revision recovery evidence are accepted.
Identity and domain	WP03 and WP04	Selected WP10 contract work	Claims/invitations are safe and Node 0 workspace is distinct from paid case.
Participant core	WP05 and WP06	Commercial configuration in WP07	Free and Complete flows are resumable, attributable and role-scoped.
Commercial and operations	WP07, WP08, WP09 and WP10	Separate repositories can progress behind pinned contracts	Synthetic A2 covers promotion, money, documents, proposals and failure recovery.
Release and cutover	WP11, WP12 and remaining WP13	Evidence collection across repos	A1/A2 pass; B1/B2 remain separately controlled human gates.

7 Programme gates

Gate	Purpose	Owner	Verifier	Failure rule
A0	Infrastructure recovery and known-revision health	Alan	Luke	Do not proceed to participant validation. Freeze live updates, retain evidence, restore the known compatible image/schema or forward-repair under the approved runbook, and keep the Project gate blocked.
A1	Synthetic core participant journey	Alan	Luke	Stop before Complete, commercial or paid-case validation. Revert/disable the failing participant feature flag or deploy the known A0 revision; preserve synthetic data only for defect diagnosis, then purge under the runbook.

Gate	Purpose	Owner	Verifier	Failure rule
A2	Synthetic commercial, promotion, billing and document journey	Alan	Luke	Disable commercial/document/provider features, retain free/safety content, prevent live money or real documents, reconcile synthetic ledger inconsistencies, and restore the A1-known revision or forward-repair as approved.
B1	Real-user and PII beta readiness	Luke	External/TBD	No real-user invitation. Continue synthetic-only, close control gaps, invalidate stale evidence and re-run the exact affected reviews/tests before a new go/no-go.
B2	PHI readiness and formal go/no-go	Luke	External security/privacy/compliance reviewers	PHI remains prohibited. Keep B1 PII scope only if still independently valid; disable any PHI-capable upload/context path, remediate, and repeat formal review.

8 Recommendation disposition matrix

The full assessment record, including evidence, assumptions, impacts, alternatives and migration implications, is in WP00. This is the decision view:

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-01	Treat PR #124 as WP00 extraction material, not a merge/conflict-resolution unit.	Accept	No independently mergeable unit survives the stale base and domain blockers. Preserve concepts/tests through clean replacement work.	WP00; P03-02/03; P04-01/02/03; P05-02/03; P06-02/03; P09-04; P10-02; P11-03/04; P12-02	No - disposition resolved; closure/deletion still requires later team confirmation.

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-02	Use corrected PR #122 as the invitation/case-shell baseline.	Accept with modification	#122 is authoritative for named tested behavior and reusable tests, not for final entities, runtime DDL, generic fact/proposal authority or the whole journey model.	WP01; WP02; WP03; WP09; WP10; WP11; WP12	No.
REC-03	Node 0 must delegate invitations to shared authority and make issuance/redemption transactionally complete.	Accept with modification	One invitation authority serves separate resource targets and lifecycles; optional profile completion is resumable rather than forced into membership redemption.	WP03 P03-02/03; WP06 P06-02; A1/A2	No; D-AUTH-02 remains for Cognito admission mechanics.
REC-04	Resolve canonical identity, participation, access and authorization before schema encoding.	Accept with modification	Identity binds principal to person; participation records domain relationship; grants carry resource capabilities; authorization uses verified context, ownership and purpose. Lens is presentation only.	P03-02/03/05; P04-01/02; P09-04; P13-05; A1/B1	Resolved as D-AUTH-03.

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-05	Restrict IP-entered pre-join GC information and remove unverified suitability ranking/gating.	Accept with modification	Allow minimum-necessary relationship/outreach/logistics context only. Health, mental health, substances, citizenship, race, pregnancy, insurance and eligibility normally wait for authenticated GC entry; no provisional value drives critical decisions.	WP04; WP05 P05-02; WP06; WP10; A1/A2	Resolved as D-DATA-03; exact cross-role fields remain D-DATA-01.
REC-06	Correct shared-memory private notes and migrate already-written rows.	Accept with modification	Run a metadata-only inventory first; record zero-row evidence if none. If rows exist, deny shared/Ask reads immediately and migrate narrowly without broadening access.	WP04 P04-02; WP10 P10-02; A1/A2	No; actual affected-row count is execution evidence, not an architecture choice.

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-07	Preserve Wave 2 work from 5b63e3a rather than reimplementing useful behavior.	Accept with modification	Treat 5b63e3a as the strongest cumulative evidence source, not an implementation baseline. Reuse tests and semantics only after the named contract changes.	WP00; WP03; WP09; WP11; WP12; WP13; A2/B1	No.
REC-08	Separate metadata-only document requests from the full content pipeline.	Accept with modification	A request/readiness marker is not a document. Content state and review begin only after a version exists, and partial merges fail closed.	WP09 P09-01/03/04/05/06; P12-02; A2	D-DOC-01/02 and D-PHI-01 remain open for content.
REC-09	Move foundational RLS and non-owner runtime before real-user PII.	Accept	App-only checks are inadequate for real-user PII given current broad readers. No alternative isolation model has equivalent evidence at lower programme cost.	WP13 P13-05/06/07; B1/B2	D-PHI-01/02 remain for approved scope/controls/approvers, not for whether RLS precedes B1.

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-10	Make #122 Python CI required, stable and resistant to bypass.	Defer	Accept the workflow recommendation now, but enforcement choice/evidence is unavailable. Until D-CI-01 resolves, do not call it bypass-resistant or open real-user gates.	WP01 P01-01; A0 and every later app PR	Yes - D-CI-01, owner Luke; options and required read-back/negative test are recorded.
REC-11	Add an explicit Node 0 IP-owner guard and correct non-IP behavior.	Accept with modification	Use one subject-derived capability policy across commands and pages; return explicit deny/role destination without creating a case.	WP03 P03-05; WP05 P05-03; A1	No.
REC-12	Bound and type Node 0 inputs and malformed request behavior.	Accept	Use strict enums/max lengths plus body/collection limits and controlled 4xx errors; do not silently truncate.	WP05; WP06; WP09; WP12	No; exact numerical limits are PR acceptance details and D-DOC-02 for files.
REC-13	Move candidate and invitation invariants from app checks into PostgreSQL migrations.	Accept	Business invariants that can be represented in PostgreSQL must remain true independently of application races.	WP02; WP03; WP04; WP05; A1	No.

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-14	Replace broad public/cloud prefixes with explicit exposure.	Accept	Explicit route identity is the simplest fail-closed control for the current app.	WP03; WP05; WP06; WP09; WP13; A1/A2	No.
REC-15	Project expiry on GET and add token/PDF cache, referrer and indexing protections.	Accept	Read projection and mutation must agree, and sensitive responses must be non-cacheable with restrictive referrer/noindex policy.	WP03; WP05	No.
REC-16	Bound candidate/invitation and memory queries and remove N+1 behavior.	Accept	Filter/limit at the authoritative storage boundary rather than after loading broad rows.	WP05; WP09; WP10; WP12	No.
REC-17	Add candidate retention, archive, correction and deletion behavior.	Accept	Implement the lifecycle in the candidate domain; keep the exact duration as the named bounded privacy decision.	WP04; WP05 P05-02; A1	Yes - D-DATA-02 supplies duration and exception policy before live use.
REC-18	Move #124's large global CSS addition to route-scoped Node 0 styling.	Accept with modification	Port only necessary styles through route scope and released DS contracts; block on interaction/accessibility/security, not cosmetic literal purity.	WP05; WP11; WP12	No.

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-19	Retain unaddressed claim semantics and one-time share-link rotation.	Accept with modification	Retain token mechanics and low-risk semantics only for a pending free-workspace claim; never layer bearer access onto a paid case.	WP03 P03-02/03; WP06	Resolved by D-AUTH-01.
REC-20	Standardize organization handling on NATALYX_ORG_ID.	Reject	Configuration unification is convenient evidence, not the final authorization model. Organization must be verified from the authenticated/granted resource context.	WP03 P03-05; WP13 P13-05	No.
REC-21	Land the GC at shared /case/home and remove a duplicate GC Node 0 home.	Accept with modification	Keep one shared participant shell pattern, but route by actual resource: free-workspace GC projection before promotion, case home only after paid-case access exists.	WP03; WP06; WP09; WP12	No.

ID	Recommendation	Disposition	Rationale	Affected work	Decision
REC-22	Keep return_to, token-scoped landing, duplicate-home cleanup, symbol cleanup and refreshed DS assets from #124.	Accept with modification	Adopt the behavior and cleanup, but keep redirect validation, resource-specific landing and canonical DS release ownership.	WP03; WP05; WP11; WP12	No.

9 Work-package details

9.1 WP00: Preservation, cloud diagnosis, and PR reconciliation

Problem. The live app is returning ALB 503 responses while overlapping draft PRs and protected dirty worktrees contain useful but contradictory implementations.

Target outcome. A source-of-truth baseline with Alan's completed review independently dispositioned, existing work mapped, and explicit no-delete/no-close conditions before replacement work begins.

Proposed lead/review. Luke leads; Alan reviews. Ready for Alan review and joint acceptance; no new implementation is authorized by this review package.

Key boundaries.

- Treat every existing branch and worktree as read-only evidence until its valuable files, tests, and reviewer findings have a named destination.
- Use Drive as the canonical review surface; the locally validated Project catalogue remains benched until separately reactivated and synced.
- Do not infer live health from GitHub build success; Gate A0 requires a known revision and live evidence.

PR	Repository	Suggested change	Depends on	Gate
P00-01	NatalyxPlatform/.github	Approve programme baseline and extraction ledger	None	Pre-A0

Acceptance focus.

- SHA and timestamp reproducibility
- Drive package/document identifier parity
- No dirty-worktree mutation
- HTTP and workflow evidence captured separately

Primary risks.

- Review findings arrive after decomposition
- A protected worktree contains the only copy of useful code
- A green PR is mistaken for live readiness

Full implementation contract: `WP00 - Preservation, cloud diagnosis, and PR reconciliation.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.2 WP01: CI and PostgreSQL evidence gate

Problem. Main builds containers and runs a deploy-contract smoke, but does not run the Python suite or prove current schema behavior against PostgreSQL.

Target outcome. A first, deliberately CI-only PR that runs the current tests, exercises the existing schema on PostgreSQL, and publishes diagnosable evidence under stable required-check names without introducing the future migration framework.

Proposed lead/review. Alan leads; Luke reviews. Technical evidence foundation.

Key boundaries.

- PostgreSQL is production authority; SQLite remains a local compatibility target, not production evidence.
- Separate current-schema evidence from tests that require the future migration mechanism.

PR	Repository	Suggested change	Depends on	Gate
P01-01	NatalyxPlatform/natalyx-app	Establish current-test and PostgreSQL CI evidence	P00-01	A0

Acceptance focus.

- Current unit suite
- Fresh PostgreSQL database
- Current init_db idempotency
- Failure artifact and log clarity

Primary risks.

- CI exposes existing backend drift
- Tests accidentally depend on protected branch code
- Secrets or DB credentials leak in logs

Full implementation contract: `WP01 - CI and PostgreSQL evidence gate.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.3 WP02: Versioned migrations and readiness

Problem. Startup calls a monolithic CREATE/ALTER routine without an ordered migration ledger, advisory lock, expand/contract contract, or reliable partial-failure recovery.

Target outcome. An approved migration ADR, versioned runner, production-safe locking/readiness behavior, baseline migration, and forward-repair procedure proven on PostgreSQL and compatible with supported SQLite development.

Proposed lead/review. Alan leads; Luke reviews. Alan chooses the migration framework/mechanism; Luke approves compatibility and risk.

Key boundaries.

- Alan evaluates Alembic versus a custom ordered runner; a custom runner is not the default without ADR evidence.
- Migrations are forward-only by default, transaction-scoped where PostgreSQL permits, advisory-locked, checksum-recorded, and expand/contract compatible with the previous app revision.

PR	Repository	Suggested change	Depends on	Gate
P02-01	NatalyxPlatform/natalyx-app	Decide migration architecture and compatibility contract	P01-01	A0
P02-02	NatalyxPlatform/natalyx-app	Introduce versioned migration runner and readiness checks	P02-01	A0
P02-03	NatalyxPlatform/natalyx-app	Baseline canonical schema and prove expand-contract repair	P02-02	A1
P02-04	NatalyxPlatform/natalyx-intelligence	Move intelligence persistence to a revisioned migration namespace	P02-01, P02-02	A1

Acceptance focus.

- Fresh and upgraded PostgreSQL
- Concurrent migrators
- Old/new app compatibility
- Interrupted migration and forward repair
- SQLite support decision

Primary risks.

- Irreversible DDL
- Concurrent deploy race
- Long backfill lock

- SQLite semantics hide PostgreSQL failure

Full implementation contract: `WP02 - Versioned migrations and readiness.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.4 WP03: Authentication, invitations, and resume

Problem. Current invitation code mixes low-risk onboarding links with case membership, and existing flows need explicit recipient binding, return-to safety, consent, expiry, revocation, duplicate-claim, and mismatch semantics.

Target outcome. Separate workspace-claim and paid-case invitation lifecycles with safe authentication/resume behavior and no link-possession path to sensitive case access.

Proposed lead/review. Alan leads; Luke reviews. Alan implements identity/admission controls; Luke reviews founder access constraints.

Key boundaries.

- An email-less link may create a pending low-risk workspace claim only.
- Identity binds an authenticated principal to a person; participation records the person's domain relationship; access grants carry resource-scoped capabilities; authorization consumes all three plus ownership and purpose.
- Paid-case membership requires authenticated identity binding, recipient or issuer confirmation as appropriate, explicit consent, and auditable redemption.
- Return targets are internal allowlisted paths and survive login/signup/confirmation without becoming redirect authority.

PR	Repository	Suggested change	Depends on	Gate
P03-01	NatalyxPlatform/natalyx-app	Harden auth return-to and participant resume	P01-01	A1
P03-02	NatalyxPlatform/natalyx-app	Add low-risk workspace claim lifecycle	P03-01, P03-05, P04-01	A1
P03-03	NatalyxPlatform/natalyx-app	Bind paid-case invitations to identity and consent	P03-01, P03-05	A2
P03-04	NatalyxPlatform/natalyx-infra	Replace shared signup admission with claim-aware Cognito admission	P03-02, P03-03	A1
P03-05	NatalyxPlatform/natalyx-app	Implement canonical identity, participation, access, and authorization contracts	P02-03, P03-01	A1

Acceptance focus.

- Logged-out resume
- Token replay/expiry/revocation
- Duplicate and concurrent claims
- Identity mismatch
- Workspace-to-case access escalation

Primary risks.

- Shared links forwarded
- Account mismatch
- Confused-deputy redirects
- Workspace data exposed during upgrade

Full implementation contract: `WP03 - Authentication, invitations, and resume.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.5 WP04: Node 0 workspace and paid-case separation

Problem. PR #124 creates a synthetic canonical case as soon as Node 0 is opened, collapsing free exploration, workspace membership, commercial commitment, and paid-case truth.

Target outcome. A distinct Node 0 workspace aggregate with attributable revisions and an idempotent, purpose-limited projection into one paid case only after the authorized commercial event.

Proposed lead/review. Alan leads; Luke reviews. Alan implements the domain/provenance contracts; Luke reviews product/privacy boundaries.

Key boundaries.

- Workspace, entitlement, commercial, payment, and case lifecycles are separate entities and state machines.
- Promotion copies only allowlisted projections with provenance, visibility, and verification intact.
- Compatibility adapters read old case-shaped Node 0 records while new writes target the workspace model.

PR	Repository	Suggested change	Depends on	Gate
P04-01	NatalyxPlatform/natalyx-app	Separate Node 0 workspace storage from paid cases	P02-03, P03-05	A1
P04-02	NatalyxPlatform/natalyx-app	Add attributable workspace revision and projection contracts	P04-01, P03-05	A1
P04-03	NatalyxPlatform/natalyx-app	Adapt legacy Node 0 case-shaped data without semantic promotion	P04-02	A1

Acceptance focus.

- One workspace/one promotion
- Retry/concurrency
- No implicit fact upgrade
- Private candidate isolation
- Old/new revision coexistence

Primary risks.

- Semantic data corruption during backfill
- Duplicate cases
- Private candidate leakage
- Older app revision writes legacy shape

Full implementation contract: `WP04 - Node 0 workspace and paid-case separation.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.6 WP05: Node 0 Free, Pathfinder, and resume

Problem. Pathfinder contains service/LLM-era routing complexity and Node 0 Free must satisfy the complete-the-match boundary without payment while preserving explicit IP authorship.

Target outcome. A deterministic entry router and resumable Node 0 Free flow for education, candidate tracking, preliminary safety checks, outreach guidance, and a willing candidate without a paid case.

Proposed lead/review. Luke leads; Alan reviews. Luke owns Free implementation; Alan reviews technical architecture, security and failure behavior.

Key boundaries.

- Routing is deterministic policy, not LLM authority.
- Candidate responses are provisional IP assertions, never GC-authored or medically verified facts.
- Safety-critical information and complete-the-match capability remain free.

PR	Repository	Suggested change	Depends on	Gate
P05-01	NatalyxPlatform/natalyx-app	Replace Pathfinder routing with deterministic entry decisions	P03-01, P04-01	A1
P05-02	NatalyxPlatform/natalyx-app	Rebuild Node 0 Free candidate and outreach domain	P04-02, P05-01	A1
P05-03	NatalyxPlatform/natalyx-app	Ship resumable Node 0 Free participant surface	P03-01, P05-02, P11-03	A1

Acceptance focus.

- All deterministic route outcomes
- Free candidate/outreach boundary
- Resume across sessions
- Critical-flag fail-safe
- Mobile interaction

Primary risks.

- Paid features leak into Free
- Provisional score is overclaimed
- Route drift
- LLM dependency reintroduced

Full implementation contract: `WP05 - Node 0 Free, Pathfinder, and resume.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.7 WP06: Node 0 Complete and GC soft onboarding

Problem. Optional paid depth, GC soft onboarding, and readiness reporting need clear entitlement gates without creating a managed case or treating IP assertions as GC answers.

Target outcome. Versioned Node 0 Complete entitlement, authenticated low-risk GC onboarding, attributable GC confirmation/replacement, and a readiness report that remains preliminary.

Proposed lead/review. Alan leads; Luke reviews. Alan implements Complete/GC onboarding; Luke reviews cross-role projection and product boundaries.

Key boundaries.

- Node 0 Complete is a flat configurable entitlement fully creditable against the JMF.
- GC confirmation appends an attributable revision; it never silently overwrites the IP source.
- GC-private answers stay private; only explicit purpose-limited projections reach the IP.

PR	Repository	Suggested change	Depends on	Gate
P06-01	NatalyxPlatform/natalyx-app	Add Node 0 Complete entitlement and versioned feature gates	P04-01, P07-01, P08-01	A2
P06-02	NatalyxPlatform/natalyx-app	Add GC soft onboarding and attributable confirmation	P03-02, P04-02, P06-01	A2
P06-03	NatalyxPlatform/natalyx-app	Add readiness report and role-scoped progress surfaces	P06-02, P10-02, P11-04	A2

Acceptance focus.

- Entitlement versioning
- GC-authored revisions
- IP/GC visibility pairs
- Revoked/expired claims
- No paid-case creation

Primary risks.

- Entitlement mistaken for case access
- GC answers exposed to IP
- IP assertions silently converted
- Price hard-coded

Full implementation contract: `WP06 - Node 0 Complete and GC soft onboarding.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.8 WP07: Commercial configuration, enrollment, and promotion

Problem. The business model requires versioned price/terms, explicit commitment, and one precise processor-neutral event before workspace-to-case promotion; none is currently authoritative.

Target outcome. Versioned commercial configuration and acceptances, a separate commitment/enrollment lifecycle, one processor-neutral byos.workspace_promotion_authorized.v1 event, and exactly-once paid-case creation by its consumer.

Proposed lead/review. Luke leads; Alan reviews. Luke implements settled commercial semantics; Alan reviews technical mechanisms and idempotency.

Key boundaries.

- Every commitment, invoice, payment, credit, refund, and promotion records the activated commercial-configuration version.
- Only the app-owned commitment/commercial domain emits byos.workspace_promotion_authorized.v1 after durable funding/terms/credit validation; a processor callback or redirect never emits it.
- The consumer creates/fetches one enrollment_pending case and restricted IP membership; GC case access follows the paid-case invitation contract.

PR	Repository	Suggested change	Depends on	Gate
P07-01	NatalyxPlatform/natalyx-app	Add versioned commercial configuration and terms acceptance	P02-03	A2
P07-02	NatalyxPlatform/natalyx-app	Implement commitment and enrollment-pending lifecycle	P07-01, P08-01	A2

PR	Repository	Suggested change	Depends on	Gate
P07-03	NatalyxPlatform/natalyx-app	Emit processor-neutral workspace promotion authorization	P04-02, P07-02, P08-02	A2
P07-04	NatalyxPlatform/natalyx-app	Promote workspace exactly once from authorization event	P03-03, P04-02, P07-03	A2

Acceptance focus.

- Terms/config version pinning
- Commitment retries
- Exact-once promotion
- Funding reversal handling
- No GC access escalation

Primary risks.

- Configuration drift
- Duplicate promotion
- Redirect spoofing
- Unresolved policy encoded as default

Full implementation contract: `WP07 - Commercial configuration, enrollment, and promotion.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.9 WP08: Payments, invoices, credits, and refunds

Problem. No append-only commercial ledger or processor-neutral payment boundary currently supports the $8,900 JMF, $750 enrollment, Node 0 credit, milestones, deterministic refunds, or failure recovery.

Target outcome. Immutable commercial ledgers, idempotent provider inbox, milestone engine, reproducible credits/refunds, reconciliation, and safe non-critical pause behavior.

Proposed lead/review. Luke leads; Alan reviews. Luke implements payment-domain contracts after D-PAY-01; external review remains mandatory before real money.

Key boundaries.

- Payments, credits, refunds, adjustments, invoices, and commercial events are append-only; mutable balances are derived projections.
- Processor events enter an idempotent inbox and become internal commands/events only after signature and amount/currency checks.
- Billing follows durable case evidence and never acts as first notification of clinical events.

PR	Repository	Suggested change	Depends on	Gate
P08-01	NatalyxPlatform/natalyx-app	Add append-only commercial and money ledgers	P02-03, P07-01	A2
P08-02	NatalyxPlatform/natalyx-app	Add processor-neutral payment command and webhook inbox	P08-01, P07-02	A2
P08-03	NatalyxPlatform/natalyx-app	Implement milestone invoicing, credits, and deterministic refunds	P08-02, P07-04, P09-04	A2
P08-04	NatalyxPlatform/natalyx-app	Add payment reconciliation and safe service-pause operations	P08-03	A2

Acceptance focus.

- Webhook replay/order
- Ledger invariants
- Milestone idempotency
- Refund table
- Node 0 credit floor
- Reconciliation mismatch

Primary risks.

- Double charge/refund
- Processor lock-in
- Clinical timing harm
- Pass-through commingling
- Policy gaps

Full implementation contract: `WP08 - Payments, invoices, credits, and refunds.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.10 WP09: Documents, active case, and provider handoffs

Problem. Current documents are metadata-only; the dirty provider branch is unreviewed and can over-project facts. There is no durable private S3/KMS/quarantine/recovery pipeline.

Target outcome. A metadata-only requirement/work-item lifecycle distinct from the secure document-content lifecycle, resilient storage/scan/download, usable active-case surfaces, and consent-scoped immutable provider snapshots with operational review.

Proposed lead/review. Alan leads; Luke reviews. Alan implements document/provider controls; Luke reviews retention, purpose and PHI boundaries.

Key boundaries.

- A requirement, readiness marker or review work item is not a document and cannot be approved as if content exists.
- Private bucket, block-public-access, KMS, opaque keys, application authorization, size/type/checksum gates, quarantine/scan, audit, retention, and orphan cleanup are mandatory for content.
- Provider handoffs are purpose-bound snapshots with checksum/revision/expiry and consent; provider-visible items are allowlisted, not inferred from broad visibility.
- ECS replacement must not lose upload/task state; recovery resumes from durable metadata and object state.

PR	Repository	Suggested change	Depends on	Gate
P09-01	NatalyxPlatform/natalyx-app	Define document lifecycle, authorization, and retention contract	P00-01, P02-01	A2
P09-02	NatalyxPlatform/natalyx-infra	Provision private encrypted document and quarantine storage	P09-01, P13-02	A2
P09-03	NatalyxPlatform/natalyx-app	Implement resilient document upload, scan, and download	P09-02, P02-03	A2
P09-04	NatalyxPlatform/natalyx-app	Ship phase-1 case progress and metadata-only request/review	P03-03, P03-05, P07-04, P09-01, P11-03	A2
P09-05	NatalyxPlatform/natalyx-app	Implement consent-scoped provider handoff snapshots and operations	P04-02, P09-06, P10-02	A2
P09-06	NatalyxPlatform/natalyx-app	Integrate content-bearing document versions into participant surfaces	P09-03, P09-04	A2

Acceptance focus.

- Unauthorized object access
- Oversize/type/spoof
- Scan failure/quarantine
- Partial upload/task loss
- Snapshot immutability/revocation
- Mobile documents

Primary risks.

- PHI exposure

- Malware
- Orphaned objects
- Stale provider packet
- Dirty branch leakage

Full implementation contract: `WP09 - Documents, active case, and provider handoffs.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.11 WP10: Intelligence, memory, proposals, and Ask safety

Problem. Private memory, shared case truth, Learn context, and AI proposals have partial contracts but current app code can place IP guide notes into shared memory and provider projections lack provenance enforcement.

Target outcome. Versioned scoped context envelopes, separate private/shared memory revisions, evidence-bound proposal approval, grounded Ask behavior, and cross-service safety evaluation/observability.

Proposed lead/review. Alan leads; Luke reviews. Alan implements intelligence contracts after D-AI-01; Luke reviews truth and privacy policy.

Key boundaries.

- The app owns case truth, authorization, PHI policy, and durable approval; intelligence returns advisory proposals only.
- Every request declares general Learn, private-user, approved-shared-case, or provider-purpose scope; contexts are not merged.
- Failures degrade explicitly with source/boundary notices and no fabricated case answer.

PR	Repository	Suggested change	Depends on	Gate
P10-01	NatalyxPlatform/natalyx-intelligence	Version the scoped intelligence context envelope	P01-01, P04-02	A2
P10-02	NatalyxPlatform/natalyx-app	Enforce private/shared memory and evidence-bound proposal approval	P04-02, P10-01	A2
P10-03	NatalyxPlatform/natalyx-intelligence	Harden Ask and Learn grounding and degraded-mode behavior	P10-01, P11-02	A2
P10-04	NatalyxPlatform/natalyx-intelligence	Add intelligence safety evaluations and cross-service observability	P10-02, P10-03	A2

Acceptance focus.

- Scope isolation
- Proposal approval/rejection
- Citation/source validity
- Unavailable service
- Prompt/context leakage
- Trace redaction

Primary risks.

- Private-to-shared leakage
- Hallucinated truth
- Sensitive logs
- Contract version drift

Full implementation contract: `WP10 - Intelligence, memory, proposals, and Ask safety.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.12 WP11: Learn release and design-system contract

Problem. Learn artifact generation and catalog work is dirty/unreleased, while participant surfaces need governed recipes without making aesthetic purity a universal release gate.

Target outcome. Validated provenance-bearing Learn artifacts, approved-only serving, release-blocking DS accessibility/security/interaction contract, and app adoption of stable recipes/source cards.

Proposed lead/review. Luke leads; Alan reviews. Luke implements the separate Learn, intelligence-catalog, participant-DS and app-adoption PRs; Alan reviews technical contracts.

Key boundaries.

- Learn resources own authored artifacts/provenance; intelligence owns approved catalog delivery; app owns authorized presentation and context.
- Security, accessibility, responsive layout, and interaction defects can block release; token/style purity alone is backlog unless tied to concrete risk.
- Participant and internal/operator design systems remain separate.

PR	Repository	Suggested change	Depends on	Gate
P11-01	NatalyxPlatform/natalyx-learn-resources	Publish validated Learn artifact and provenance contract	P00-01	A2
P11-02	NatalyxPlatform/natalyx-intelligence	Serve approved Learn catalog versions from intelligence	P11-01	A2
P11-03	NatalyxPlatform/natalyx-design-system	Release participant design-system accessibility and interaction contract	P00-01	A1

PR	Repository	Suggested change	Depends on	Gate
P11-04	NatalyxPlatform/natalyx-app	Adopt approved Learn and source-card recipes in app	P11-02, P11-03	A2

Acceptance focus.

- Artifact schema/provenance
- Approved-only catalog
- Missing-resource empty state
- Recipe escaping
- Keyboard/reduced-motion/mobile

Primary risks.

- Fake/unapproved content
- Cross-repo version drift
- Aesthetic review blocks recovery
- Internal DS leakage

Full implementation contract: `WP11 - Learn release and design-system contract.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.13 WP12: Ask/Learn shell and mobile cutover

Problem. Participant Ask/Learn entry points, case shell, legacy routes, and mobile interactions are not yet one coherent clickable journey with explicit context boundaries.

Target outcome. A unified scope-labelled Ask/Learn shell, accessible mobile participant journey, and verified redirect-based legacy cutover with no broken or misleading controls.

Proposed lead/review. Luke leads; Alan reviews. Luke implements participant shell/cutover; Alan reviews integration, authorization and failure behavior.

Key boundaries.

- The shell labels general, private, and shared-case context and never implies a source exists when it does not.
- Server owns state and permissions; client code enhances interaction only.
- Legacy routes redirect through allowlisted, telemetry-observed compatibility paths before deletion.

PR	Repository	Suggested change	Depends on	Gate
P12-01	NatalyxPlatform/natalyx-app	Unify Ask and Learn shell with explicit context scopes	P10-03, P11-04, P09-04	A2
P12-02	NatalyxPlatform/natalyx-app	Complete mobile participant interaction and accessibility cutover	P05-03, P06-03, P09-04, P09-06, P12-01	A2
P12-03	NatalyxPlatform/natalyx-app	Retire legacy participant routes behind verified redirects	P12-02, P13-04	B1

Acceptance focus.

- Clickable IP/GC flows
- Context labels
- No-JS/reduced-motion/keyboard
- Mobile widths
- Redirect and stale bookmark
- Logout/login/resume

Primary risks.

- Context ambiguity
- Mobile-only dead ends
- Premature legacy removal
- Design drift

Full implementation contract: `WP12 - Ask and Learn shell and mobile cutover.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

9.14 WP13: Deployment, cloud gates, and PHI readiness

Problem. The dev endpoint is presently 503, deployments do not prove application readiness, and test/deploy success cannot establish real-user or PHI readiness.

Target outcome. Known-revision diagnostics, guarded rollback/evidence capture, automated synthetic A1/A2 journeys, and explicit B1/B2 operational/privacy/PHI go/no-go packets.

Proposed lead/review. Alan leads; Luke reviews. Alan owns technical gate evidence; Luke accepts gates and external reviewers retain their required attestations.

Key boundaries.

- Health, readiness, revision, dependency, migration, and safe configuration signals are distinct.
- Every cloud gate has prerequisites, evidence, owner, verifier, and failure/rollback response.
- B2 requires formal security/privacy/compliance evidence and human go/no-go; it is not a software test result.

PR	Repository	Suggested change	Depends on	Gate
P13-01	NatalyxPlatform/natalyx-app	Expose known-revision health, readiness, and dependency diagnostics	P01-01	A0
P13-02	NatalyxPlatform/natalyx-infra	Harden guarded deployment, rollback, and evidence capture	P01-01, P13-01	A0
P13-03	NatalyxPlatform/natalyx-app	Automate synthetic A1 core participant journey verification	P03-02, P04-03, P05-03, P11-03, P13-02	A1
P13-04	NatalyxPlatform/natalyx-app	Automate synthetic A2 commercial and document journey verification	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03	A2
P13-05	NatalyxPlatform/natalyx-app	Enforce case RLS and a non-owner PostgreSQL runtime	P02-03, P03-05, P04-03, P08-04, P09-05, P10-02	B1
P13-06	NatalyxPlatform/natalyx-infra	Harden B1/B2 backups, audit, alarms, and cohort controls	P09-02, P13-02, P13-05	B1
P13-07	NatalyxPlatform/.github	Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks	P13-04, P13-05, P13-06	B1

Acceptance focus.

- Known revision
- ALB/ECS/dependency health
- Rollback
- Synthetic A1/A2
- PII controls
- PHI go/no-go evidence

Primary risks.

- Unknown running revision
- Rollback incompatible with schema
- Synthetic success overclaimed
- PHI enabled without organizational controls

Full implementation contract: `WP13 - Deployment, cloud gates, and PHI readiness.md` in `MD Work Packages`; matching review PDF in `PDF Work Packages`.

10 Founder decision resolution and remaining controls

Founder-final policy is not the same as an active technical control. The status, remaining owner and programme effect below preserve that distinction.

Decision	Status	Founder outcome	Remaining owner	Resolve by	Programme effect
D-MIG-01	Founder constraints final; Alan immediate; operationally pending	Reliability-first migrations with zero intended data loss; PostgreSQL is authoritative for production and SQLite may remain a convenient synthetic local-development projection without full production parity.	Alan - immediate	Before P02-02	P02-01 decision work may proceed; P02-02, P02-03 and Gate A0 remain blocked until Alan's mechanism and evidence are accepted.
D-AUTH-01	Resolved by reconciliation	A bearer link creates only a pending low-risk workspace claim; paid-case access requires authenticated binding, explicit consent and recipient/issuer confirmation.	Alan - implementation	Resolved 2026-07-12	No longer blocks P03-02 or P03-03.
D-AUTH-02	Founder constraints final; Alan later	Participant admission uses short-lived signed or server-verified proof bound to email, purpose, claim and expiry; it admits account creation only and never grants workspace or case membership. Internal admission remains a separate allowlist path.	Alan - later at P03-04	Before P03-04	P03-04 and Gate A1 remain blocked until the technical mechanism and evidence exist.
D-AUTH-03	Resolved by reconciliation	Identity, participation, access grants, ownership and operation authorization remain separate; participation or lens alone never authorizes.	Alan - implementation	Resolved 2026-07-12	No longer blocks the reconciled access/workspace PR sequence.

Decision	Status	Founder outcome	Remaining owner	Resolve by	Programme effect
D-CI-01	Founder final; operationally pending	Option A - GitHub Team approved as the private-repository enforcement control.	Alan - immediate	Founder decision resolved 2026-07-13; control required before P01-01 merge or any later app implementation PR	P01-01 merge, Gate A0 and dependent implementation PRs remain blocked until the live control is active and verified.
D-DATA-01	Founder decision final; implementation evidence pending	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.	Alan - implementation	Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-DATA-02	Founder default final; external validation before B1	Retain candidate/workspace material for a 90-day minimum after explicit rejection, withdrawal or last legitimate outreach, honor earlier valid deletion, retain only non-identifying funnel metrics and a minimal audit tombstone, and preserve legal holds. Corrections append source-labelled revisions and private IP notes remain author-only.	Alan - implementation; privacy/legal - validation	Before P04-02	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
D-DATA-03	Resolved by reconciliation	No candidate ranking, viability/eligibility flag or race preference survives; provisional pre-join input can generate neutral source-labelled questions only.	Alan - implementation	Resolved 2026-07-12	No longer blocks Node 0 implementation planning.

Decision	Status	Founder outcome	Remaining owner	Resolve by	Programme effect
D-COM-01	Founder decision final for beta; tax externally dependent	Node 0 Complete is a voluntary $495 USD one-time upgrade paid by the IP or authorized IP account holder, never the GC, collected before Complete features and fully credited against the later Phase 1/JMF without gating matching or safety support.	Alan - implementation	Before activation of a Complete configuration	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Founder decision final for v1; live activation gated	Configuration v1 is $8,900 USD: a $750 deposit followed by $1,630, $2,445, $1,630, $1,630 and $815 milestones. Pay-in-full is $7,830 total, leaving $7,080 after the deposit; the separate $495 Complete credit must not be double-counted.	Alan - implementation	Before P07-01 activation data	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
D-COM-03	Externally dependent; founder policy inputs settled	No live acceptance or charge may rely on the contradictory current terms draft. D-COM-06 and D-COM-07 supply the approved business policy for a new versioned terms package.	External legal/privacy/finance; Luke final business approval	Before P07-02 user acceptance	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
D-COM-04	Founder decision final; implementation evidence pending	Use two-step commitment: settled $750 plus current accepted terms proves commercial enrollment, but payment alone never promotes. Exactly one enrollment-pending case follows explicit IP promotion and authenticated GC consent. The $1,630 activation tranche is earned only after defined IP/GC onboarding and screening coordination has begun; it never implies medical clearance.	Alan - implementation	Before P07-02	P07-02 through P09-04 may proceed under the settled semantics.

Decision	Status	Founder outcome	Remaining owner	Resolve by	Programme effect
D-PAY-01	Founder constraints final; Alan later	Offer card and US ACH, encourage but do not require ACH above $1,000, absorb ordinary beta fees without surcharging, and keep processor metadata free of GC, journey, clinical or document detail.	Alan - later before P08-02	Before P08-02 adapter	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
D-COM-05	Founder decision final for beta	Financing is excluded and hidden from beta; Natalyx does not advertise, recommend or integrate a lender or treat lender status as funding evidence.	None during beta	Before financed option is exposed	Financing no longer blocks P08-02; the feature is deliberately out of scope.
D-COM-06	Founder policy final; legal/accounting externally dependent	After post-heartbeat loss, never charge the $815 birth tranche; retain the delivered $1,630 pregnancy-support tranche and grant an equal $1,630 future credit to the same IP household for 36 months, with compassionate manual extension and defined offboarding support.	Alan - implementation; legal/accounting - wording	Before automated pregnancy-stage refund logic	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
D-COM-07	Founder policy final; legal/accounting externally dependent	Protect delivered work: the $750 is refundable until onboarding is delivered and then earned; activation follows D-COM-04. Non-clearance or pre-clearance GC withdrawal stops later charges. Unearned prepayment is refunded within ten business days, with refund or rollover choice and one replacement within 12 months without a second $750 deposit.	Alan - implementation; legal/accounting - wording	Before real billing	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
D-COM-08	Founder decision final; implementation evidence pending	Use a humane 30-day grace period with reminders seven days before due, on the due date, and at 7, 14 and 30 days overdue; no beta late fee and human outreach before pause. Only future non-critical paid work may pause after 30 days.	Alan - implementation	Before P08-04	P08-04 may proceed under the settled policy.

Decision	Status	Founder outcome	Remaining owner	Resolve by	Programme effect
D-DOC-01	Founder default final; Alan later; external validation before B1/B2	Retain clean active-case documents while active plus two years after close; signed consents, payment/access audits and deletion tombstones for seven years; failed/quarantine orphans seven days; superseded content 90 days unless required; honor valid deletion within 30 days subject to legal hold. Downloads require fresh authorization and links expire within five minutes.	Alan - later before P09-03; privacy/legal - validation	Before P09-03	P09-01 ADR work may proceed; P09-03/P09-06 and real-content activation wait for Alan's delivery choice and evidence.
D-DOC-02	Founder constraints final; Alan later	Allow PDF, JPEG, PNG and non-macro DOCX up to 50 MiB. Reject spoofed, macro-enabled and archive-like input; quarantine until clean; treat password-protected, skipped, error and timeout results as unavailable.	Alan - later before P09-02	Before P09-02	P09-01 contract work may proceed; P09-02/P09-03/P09-06 wait for Alan's technical selection and evidence.
D-PROV-01	Founder decision final; Alan later	Provider handoff is a purpose-bound immutable snapshot for one verified recipient and named purpose, with participant preview/consent, clean documents and appropriately verified source-labelled facts. Unopened access expires after 14 days and can be revoked before access; downloaded copies cannot be recalled.	Alan - later before P09-05	Before P09-05	Synthetic provider-snapshot design may proceed; P09-05 completion and real handoff require Alan's evidence and later external review.

Decision	Status	Founder outcome	Remaining owner	Resolve by	Programme effect
D-AI-01	Founder scope/cost final; Alan later; PHI externally gated	AI is bounded and advisory: role-scoped education, retrieval, summaries, question drafts, next steps and provider preparation using approved Learn and authorized source-labelled context. It never diagnoses, ranks, approves, clears or writes durable truth; authorized human approval precedes memory, case or provider material.	Alan - later before P10-01	Before P10-01	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
D-OPS-01	Founder constraints final; Alan immediate; operationally pending	Use fail-closed recovery: protect data integrity over uptime, restore only a reviewed proven-compatible application-image/schema pair, use safe read-only or unavailability when uncertain, and forward-repair when rollback is unsafe. Target 30-minute beta recovery; allow up to two hours emergency downtime and notify Luke when more than one hour is expected.	Alan - immediate	Before Gate A0 recovery action	P13-02 and Gate A0 remain blocked until the runbook and synthetic exercise pass; A1 must exercise the control.
D-PHI-01	Founder beta scope final; externally gated	B1 is an invitation-only US beta of at most five IP/GC pairs using names, verified account/contact data, relationship/outreach/logistics, consent versions, nonclinical progress, feedback and processor references only. Medical, reproductive-health, psychological, insurance-health and clinical-clearance data are prohibited; no real document bytes or provider packets; AI receives de-identified nonclinical context only.	Alan - B1/B2 evidence; external reviewers - approval	Before B1 for PII scope; before B2 for PHI	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.

Decision	Status	Founder outcome	Remaining owner	Resolve by	Programme effect
D-PHI-02	Founder governance final; external reviewers pending	Luke accepts B1 from Alan's technical evidence plus external privacy/legal advice, without separate independent technical sign-off. B1 evidence is at most 60 days old and reruns after material change or 180 days. B2 requires qualified external security, privacy/compliance and legal/vendor review with release evidence at most 30 days old and assessments at most 12 months old.	Alan - evidence; external reviewers - attestations; Luke - go/no-go	Before B1	B1 and B2 remain blocked until the corresponding evidence is current and the required humans sign off.

10.1 Immediate Alan decisions

Decision	Alan owns	Evidence	Latest safe point
D-CI-01	Activate GitHub Team when enforcement is required; Configure required unittest and PostgreSQL checks; Restrict routine bypass; Supply ruleset readback and negative-bypass evidence	GitHub Team activation/readback; Mandatory unittest and PostgreSQL checks; Restricted routine bypass; Negative-bypass evidence	Founder decision resolved 2026-07-13; control required before P01-01 merge or any later app implementation PR
D-MIG-01	Choose the migration framework; Choose sequencing/locking and expand-contract mechanics; Choose the detailed rollback, restore or forward-repair mechanism	Migration ADR; Fresh/current/upgrade compatibility matrix; Tested recovery or rollback evidence; Destructive-change approval path	Before P02-02
D-OPS-01	Choose exact compatibility, rollback/forward-repair and automation mechanisms; Name the evidence owner and write the runbook	Documented recovery procedure; Successful synthetic recovery exercise; Known-compatible image/schema evidence	Before Gate A0 recovery action

10.2 Later Alan decisions

Decision	Alan owns	Evidence	Checkpoint
D-AUTH-02	Choose the proof format, signer/verifier, key lifecycle and claim lifecycle	Expiry, replay, wrong-email, wrong-purpose, mismatch and revocation tests; Separate internal allowlist evidence	Before P03-04
D-PAY-01	Choose the processor/vendor, tokenization, webhook, settlement, return/refund and reconciliation mechanisms	Provider security and cost comparison; Synthetic settlement, return, refund, duplicate and reconciliation tests	Before P08-02 adapter
D-DOC-01	Choose proxy versus presigned delivery by class; Implement hold, deletion, tombstone, retry and authorization behavior	Fresh-authorization and five-minute-link tests; Retention/hold/deletion exercise; Privacy/legal approval before real content	Before P09-03
D-DOC-02	Choose the AWS-native scanner, queue, retry, timeout, recovery and cost-control mechanisms	Type/sniffing/spoof tests; Quarantine, error, timeout and retry tests; Cost estimate and alerts	Before P09-02
D-PROV-01	Choose recipient verification, snapshot checksum, expiry, revocation and audit mechanisms	Wrong-recipient, expiry, revoke-before-access, reissue and checksum tests	Before P09-05
D-AI-01	Choose Bedrock model, prompts, retrieval, schemas, fallback, human-review rails and cost controls	Scope/authority negative tests; Fallback and malformed-response tests; Safety and cost evidence	Before P10-01
D-PHI-01	Implement inventory, cohort controls, RLS, cross-tenant denials, de-identification and prohibited-data tests	Five-pair cohort configuration; RLS/cross-tenant/de-identification/prohibited-data tests; External privacy/legal approval	Before B1 for PII scope; before B2 for PHI
D-PHI-02	Maintain evidence recency, change-triggered reruns and exception records	B1/B2 evidence index with dates; Named reviewers; Exception register; Signed go/no-go	Before B1

10.3 External dependencies

Decision	External dependency	Activation boundary
D-DATA-02	Privacy/legal validation before B1	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
D-COM-01	Tax treatment and customer-facing wording before live activation	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Approved terms and tax treatment before live activation	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
D-COM-03	Legal, privacy, tax and finance review before real-user acceptance or charging	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
D-PAY-01	Payment/security/finance review before real money	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
D-COM-06	Legal and accounting review before live use	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
D-COM-07	Legal and accounting review before live use	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
D-DOC-01	Privacy/legal validation before B1/B2 real-content use	P09-01 ADR work may proceed; P09-03/P09-06 and real-content activation wait for Alan's delivery choice and evidence.
D-PROV-01	Privacy/provider review before real handoff	Synthetic provider-snapshot design may proceed; P09-05 completion and real handoff require Alan's evidence and later external review.
D-AI-01	B2 external security/privacy/compliance approval before PHI	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
D-PHI-01	External privacy/legal approval before B1; Qualified external security/privacy/compliance/legal approval before B2	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.
D-PHI-02	Named privacy/legal advisers for B1; Named qualified security/privacy/compliance/legal reviewers for B2	B1 and B2 remain blocked until the corresponding evidence is current and the required humans sign off.

11 Handoff checklist

- All 24 decision IDs are present and uniquely classified.
- D-CI-01, D-MIG-01 and D-OPS-01 remain immediate Alan controls and keep Gate A0 blocked until evidence is accepted.
- Later Alan decisions are attached to their actual work-package checkpoints rather than forced prematurely.
- External terms, tax, privacy, security, compliance and PHI dependencies remain explicit activation gates.
- A0/A1/A2 remain synthetic; B1 is limited to at most five IP/GC pairs and approved nonclinical PII; real document bytes, provider packets and PHI remain prohibited until their later gates.
- The 52 suggested implementation PRs remain proposals; this planning update creates, merges or deploys none of them.